Activity - 1

1) Medical Image Enhancement

(i) The best gray-level transformat
for this is log transformation

(ii) Log transformation is best because
expands the dark pixel values &
compresses the brighter pixel valu
It helps improve the visibility o
darker tissues

(iii) After using this, the darker
tissues become more visible,
while the brighter bones remain
visible with reduced intensity
differences.

2) Night time surveillance:

(i) For this, Power - Law (Gamma)
transformation with ($\gamma < 1$)

(ii) It brightens dark regions
while preserving the natural
appearance of scene. Negative
transformation reverses brightn
& may make the image look
unnatural.

(iii) The parking lot becomes
brighter, & objects such as
vehicles people & other details

(3) Fingerprint Recognition:

(i) for this, contrast stretching.

(ii) Contrast stretching is better option because it expands the narrow range of gray levels, increasing the diff b/w fingerprint ridges & valleys. This makes the edge & ridges better

(iii) log transformation would compress the darker pixels & brighter pixel values. It might improve some details, but it may not enhance the contrast b/w ridges & valleys as effectively as contrast stretching.